

What Do Relation Labels Add to Image–Text Retrieval?

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Abstract

An image search system should reject wrong descriptions while recognizing different correct descriptions. Labels can mark a caption as supported, contradicted, or unresolved by an image. We ask what each use of these labels adds to retrieval. Twelve training rules use one pretrained model’s fixed features, with matched caption pools and tuning budgets. Compared with training on the original captions, adding supported captions improves the annotated statement test by 5.81 percentage points. It gains only 0.14 points on edited captions and loses 8.56 points when retrieving captions for images. Gains concentrate on added objects and attributes; accuracy falls on replacement and swap edits. Extra penalties for contradicted captions and omitting unresolved pairs add no clear further gain. Uniformly reducing the influence of competing captions preserves more retrieval accuracy, but scores lower on the annotated task. Relation annotation must earn its value through the matches a search system returns.

1 Introduction

An image search system must distinguish an incorrect description from a different correct description. Consider a photo of a child holding a red ball. Both “a child with a red ball” and “a red ball held by a child” can retrieve it. “A child with a blue ball” should not. A statement such as “the child owns the ball” is unresolved by the image. These distinctions motivate three caption labels: *supported*, *contradicted*, and *neutral*.

CLIP learns image and text vectors by scoring associated image–caption pairs above other candidates (Radford et al., 2021). Training calls the associated captions *positives* and the competing candidates *negatives*. A negative in this objective need not be a false description. It may be another valid caption, or one that the image cannot resolve. Relation labels offer a direct way to distinguish

these cases. A natural policy adds supported captions to the positive set, penalizes contradicted captions more strongly, and omits neutral pairs.

The practical question is what each use of the label buys. Extra supported captions change which descriptions the model is rewarded for matching. Contradiction weights change how strongly it separates incorrect descriptions. Neutral exclusion changes which candidates compete at all. A comparison with ordinary adaptation combines these effects. To identify the value of the relation-specific decisions, the candidate pool and the training budget must stay fixed while these operations are varied separately.

Prior work establishes why those controls should include retained matching ability. Kamath et al. (2024) show that improved rejection of hard negatives can hide failures on valid rewordings, even after generic positive augmentation. FSC-CLIP studies preservation of broader multimodal capabilities (Oh et al., 2024). CLIC and concept-centric training report joint improvements in compositionality and retained capabilities (Peleg et al., 2025; Pham et al., 2026). The retrieval cost of a particular label policy is therefore an empirical question.

The individual operations also have precedents. MAFA compares converting false negatives into positives with removing them (Byun et al., 2024). Similarity-regulated learning weights negative pairs (Jiang et al., 2023); semantics-aware triage uses similarity to divide medical image–report pairs into positive, negative, and neutral groups (Liu et al., 2023). Here the labels explicitly distinguish supported, contradicted, and unresolved statements. Our contribution is to separate their training roles in a common candidate pool, then measure which roles improve the selected search model.

We study that decision with one annotated caption pool and twelve training policies. The encoders stay fixed; every policy learns small correc-

tions to the image and text vectors. We evaluate the complete selected procedures on the annotated distinction, edited captions, valid alternative wordings, and full-pool retrieval. This connects a controlled decomposition of label use to the two outcomes needed by a search system: reject a wrong description and still rank a correct match first.

2 A Matched Study of Label Use

Same candidates, different training roles. Every image has five original source captions and annotated statements from e-SNLI-VE (Kayser et al., 2021). A batch contains its images and all their source captions and statements. *Source positives* uses the original captions as positive targets. All statements, including supported statements, are competing negatives. *Positive expansion* promotes the supported statements already in this pool. It changes their training role while keeping the images and caption strings fixed.

Starting from the expanded positive set, *contradiction weighting* doubles the weight of annotated contradictions. *Neutral exclusion* assigns zero weight to annotated neutral pairs. *Both* applies these changes together. An unannotated image-caption pair remains an eligible negative. In particular, a caption belonging to another image in the batch is not automatically neutral. The zero weight removes a known neutral pair from the objective rather than treating it as a weak positive.

Controls for the label-specific choices. Count-matched random exclusion tests whether removing specifically neutral pairs helps beyond removing the same number of negatives. Constant weighting assigns every negative weight 0.25, testing a general reduction in competition without choosing individual pairs. Two text-similarity policies use either a fixed or a batch-median threshold on frozen source-caption features; a shuffled control breaks the pair-to-weight assignments. Target smoothing and within-image supported-contradicted ranking complete the twelve policies. The comparison holds model capacity and tuning budget fixed. Appendix A defines each rule.

Training and selection. We use frozen OpenCLIP ViT-B/32 features from the LAION-2B checkpoint. Two zero-initialized residual maps add learned linear corrections to the features, with 524,288 trainable parameters; the encoders and score scale remain fixed. The image-disjoint split

contains 1,200 training, 100 calibration, 100 validation, and 400 test images.

Each configuration runs for ten epochs at three learning rates and three seeds, giving 108 runs. A fixed validation score averages retrieval accuracy at rank one and supported-versus-contradicted statement accuracy. Validation retrieval counts source or supported captions as correct. Selection includes epoch zero. One learning rate per policy is chosen across all three seeds, and each seed retains its best validation epoch. We then evaluate the resulting 36 checkpoints and the frozen reference. Appendix C specifies the recorded protocol.

Measure the annotation benefit and the search outcome. The e-SNLI-VE test measures how often a supported statement scores above a contradicted one, averaged by image. SugarCrepes tests an original correct caption against an edited wrong caption, or *foil*, on all 7,511 official pairs (Hsieh et al., 2023). These are the two primary discrimination endpoints. Six planned contrasts compare the combined policy with positive expansion and with each component removed. We use 10,000 paired image resamples and a six-comparison Bonferroni adjustment.

Two further tests measure retained matching ability. SugarCrepes++ provides two valid captions and one foil for each of 4,757 triplets. Success requires both positives to beat the foil (Dumpala et al., 2024). COCO retrieval searches all 25,000 captions for each of 5,000 images, and all images for each caption. Recall at rank one ($R@1$) is the percentage of queries whose top candidate is an annotated match. We report image-to-text ($I \rightarrow T$) and text-to-image ($T \rightarrow I$) separately. These measurements distinguish performance on a few designed alternatives from returning a correct result in a larger pool.

3 What the Labels Buy

The largest benefit is on the annotated distinction. Positive expansion improves e-SNLI-VE by 5.81 percentage points over source positives. Its SugarCrepes gain is 0.14 points. Table 1 separates the two negative-pair interventions on top of that expanded set. The combined policy differs from positive expansion by -0.26 points on e-SNLI-VE and -0.16 on SugarCrepes. Their adjusted 99.17% image-bootstrap intervals are $[-0.60, 0.07]$ and $[-0.51, 0.21]$.

Policy	e-SNLI-VE	SugarCrepe	SC++ both	COCO I→T	COCO T→I
Frozen	79.47	81.21	68.95	56.32	39.35
Source positives	81.18 $\pm$ 0.14	83.26 $\pm$ 0.30	67.40	55.85	39.38
Positive expansion	87.00 $\pm$ 0.40	83.39 $\pm$ 0.15	67.23	47.29	34.84
+ contradiction weighting	87.28 $\pm$ 0.33	83.33 $\pm$ 0.16	67.30	47.25	34.81
+ neutral exclusion	86.66 $\pm$ 0.44	83.39 $\pm$ 0.27	67.66	48.01	35.39
+ both	86.74 $\pm$ 0.17	83.24 $\pm$ 0.31	67.61	47.26	35.27
Constant weights	82.15 $\pm$ 0.24	83.67 $\pm$ 0.20	68.37	55.21	39.39

Table 1: Relation interventions with the same candidate pool and tuning budget. Scores are percentages; subscripts show sample standard deviation across three selected seeds for the primary endpoints. COCO columns report full-pool R@1. The three rows marked + add rules to positive expansion. Constant weights use source positives.

Edit group	<i>n</i>	Source	Expand.	Δ
Added content	2,754	88.09	90.01	+1.92
Replace/swap	4,757	80.46	79.56	-0.90
All edits	7,511	83.26	83.39	+0.14

Table 2: SugarCrepe accuracy by edit group. Differences are positive expansion minus source positives, in percentage points. Groups partition the official pairs.



Source positives: A young girl is preparing to blow out her candle.

Positive expansion: a woman is making some food in a bowl

Figure 1: Top retrieved captions for the same image query (seed 17). The best annotated caption moves from rank 1 to 6. Both models search the full caption pool.

The edited-caption gain targets added content. SugarCrepe mixes edits that add objects or attributes with edits that replace content or swap its assignment. The aggregate score hides opposite effects. Positive expansion improves the added-content group and reduces accuracy on the combined replacement/swap group (Table 2). Both directions hold in every seed. Appendix F reports the seven original categories.

Full-pool retrieval exposes the practical cost. Positive expansion loses 8.56 points of image-to-text R@1 relative to source positives. On the fixed pool, that is 428 fewer image queries returning an annotated correct caption first per 5,000 queries, averaged across seeds. Every seed loses retrieval accuracy. Text-to-image R@1 falls by 4.54 points as well. These tests keep all candidate captions and images in the pool, so the loss concerns the rank of a usable match rather than a binary choice against one designed foil.

Figure 1 shows a changed top result for the same image and caption pool. A lower score for a deliberately edited foil would not detect this change in the match returned to the user.

A simple control preserves a different set of capabilities. Constant negative weighting retains 7.91 points more image-to-text R@1 than positive

expansion. It also scores higher on edited captions, while its annotated statement accuracy is lower (Table 1). This control weakens all negative competition without deciding which pairs are contradicted or neutral. The selected means expose the application choice: improving the annotation distinction is different from preserving edited-caption and retrieval performance. Table 3 later compares the similarity-based weighting rules.

The use of neutral labels has a conditional benefit. Neutral exclusion beats count-matched random removal by 0.40 points on e-SNLI-VE and 1.77 on SugarCrepe. However, when contradiction weighting is retained, adding neutral exclusion changes e-SNLI-VE by -0.53 points, with adjusted interval $[-0.89, -0.19]$. Selecting neutral pairs is therefore useful relative to random removal, but it does not improve the combined selected procedure over contradiction weighting alone.

Alternative wordings test a separate requirement. All 31 nonzero-epoch adapters reduce accuracy on the alternative valid SugarCrep++ caption relative to the frozen model. This occurs across policies, including source positives. Table 1 reports the stricter score that requires both valid captions to beat the foil. Appendix E shows the paired first-caption and alternative-caption outcomes for every selected adapter.

4 Why More Positives Change the Target

The contrastive policies with uniform positive targets use a common objective. For image i and caption j , let s_{ij} be their scaled cosine score. Let P_i be the nonempty set of positive captions and $w_{ij} \geq 0$ the candidate weight. Positive weights are one; a zero weight omits a pair. For the unsmoothed policies, the image-row loss is

$$\ell_i = \log \left(\sum_j w_{ij} e^{s_{ij}} \right) - \frac{1}{|P_i|} \sum_{j \in P_i} s_{ij}. \quad (1)$$

We average it with the corresponding text-to-image loss over captions having a positive image. This mean-log-probability form gives equal target probability to every positive (Khosla et al., 2020). The exact reverse-direction normalization, smoothing, and separate ranking loss appear in Appendix A.

Positive expansion redistributes that fixed probability budget. Suppose there are K original positives and M already-present candidates become additional positives. Use the unsmoothed objective and hold the scores, finite candidate set, and unit weights fixed. For any original positive p , the derivative changes by

$$\frac{\partial \ell_{\text{expanded}}}{\partial s_{ip}} - \frac{\partial \ell_{\text{source}}}{\partial s_{ip}} = \frac{M}{K(K+M)}. \quad (2)$$

For example, adding five supported captions to five original positives halves each original target from 0.2 to 0.1. The denominator has not changed because the candidates were already present. Each added positive gains target mass; each original positive loses some. Thus correct labels can change how the objective distributes attraction among correct captions.

For each candidate, dividing its weighted exponential score by the sum inside the logarithm in Equation 1 gives its normalized probability. With two positives and one negative, probabilities (0.8, 0.1, 0.1) give the first positive derivative

Weights	e-SNLI-VE	Sugar	I→T
Fixed similarity	79.56	81.80	54.33
Shuffled weights	83.34	83.49	52.27
Median similarity	82.66	83.73	53.97

Table 3: Similarity weights and their shuffle, using source positives. Entries are mean percentages; the final column is COCO R@1. The constant-weight reference appears in Table 1.

$0.8 - 0.5 = 0.3$. An independent gradient descent step on that score would lower it even though the caption is correct. Equal targets put downward pressure on this stronger match and upward pressure on the positive below its target. Appendix I gives the derivations, the change in text-to-image averaging, and the summed-positive alternative. Target redistribution motivates retaining the source-positive comparison when evaluating extra labels.

Pair assignments and overall competition. The weighting controls separate a semantic assignment from a general change in competition (Table 3). Shuffling the fixed-threshold weights improves both primary endpoints. Median weighting is close to constant suppression on edited captions, while constant suppression retains more retrieval accuracy. The score of the weighting rule therefore depends on more than whether its weights use text similarity. Appendix I formalizes the distinction with a row-specific constant that preserves total negative mass while changing its allocation.

5 Implications for Search Evaluation

Choose a relation-supervised procedure against an acceptable retrieval loss on representative development queries. The selection score here already rewards retrieval, but on a small pool from the annotation source. Development queries should therefore reflect the deployment source and candidate-pool size. Reporting the annotated distinction, edited-caption discrimination, and the two retrieval directions separately makes the choice visible. The evaluation should also include multiple valid wordings, since a correct result must survive changes in how a user describes it.

Limitations

This study concerns one frozen OpenCLIP checkpoint and linear residual adaptation on a subset of e-SNLI-VE. It does not establish the behavior of full-encoder training, other data scales, or other model families. The relation labels partly inherit text-inference judgments; the release does not provide independent visual verification for every pair. Caption ownership also supplies incomplete retrieval relevance.

The multi-positive objective averages positive log probabilities. The summed-positive alternative in Appendix I is an analytical comparison. Independent policy selection changes the learning rate and retained epoch, so the comparisons estimate selected training procedures. Image-bootstrap intervals condition on three seeds and do not quantify uncertainty over new training runs. Sugar-Crepe++, retrieval, weighting controls, and edit-group summaries are secondary or descriptive. COCO-derived benchmarks overlap, so they measure different capabilities rather than independent replications. Appendix B records source and execution details.

Ethical Considerations and Assistance

We use existing public datasets under their source terms and collect no new human judgments. Captions and pretrained features may reflect source biases. Generative AI assisted planning, implementation, analysis, literature discovery, and drafting. The authors are responsible for checking the released computations and manuscript.

References

Jaeseok Byun, Dohoon Kim, and Taesup Moon. 2024. [MAFA: Managing false negatives for vision-language pre-training](#). In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 27314–27324.

Sri Harsha Dumpala, Aman Jaiswal, Chandramouli Sastri, Evangelos Milios, Sageev Oore, and Hassan Sajjad. 2024. [SUGARCREPE++ dataset: Vision-language model sensitivity to semantic and lexical alterations](#). *Preprint*, arXiv:2406.11171.

Cheng-Yu Hsieh, Jieyu Zhang, Zixian Ma, Anirudha Kembhavi, and Ranjay Krishna. 2023. [SugarCrepe: Fixing hackable benchmarks for vision-language compositionality](#). In *Advances in Neural Information Processing Systems*, volume 36.

Chaoya Jiang, Wei Ye, Haiyang Xu, Songfang Huang, Fei Huang, and Shikun Zhang. 2023. [Vision language pre-training by contrastive learning with cross-modal similarity regulation](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 14660–14679. Association for Computational Linguistics.

Amita Kamath, Cheng-Yu Hsieh, Kai-Wei Chang, and Ranjay Krishna. 2024. [The hard positive truth about vision-language compositionality](#). In *European Conference on Computer Vision*.

Maxime Kayser, Oana-Maria Camburu, Leonard Salewski, Cornelius Emde, Virginie Do, Zeynep Akata, and Thomas Lukasiewicz. 2021. [e-ViL: A dataset and benchmark for natural language explanations in vision-language tasks](#). In *Proceedings of the IEEE/CVF International Conference on Computer Vision*.

Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan. 2020. [Supervised contrastive learning](#). In *Advances in Neural Information Processing Systems*, volume 33.

Bo Liu, Donghuan Lu, Dong Wei, Xian Wu, Yan Wang, Yu Zhang, and Yefeng Zheng. 2023. [Improving medical vision-language contrastive pretraining with semantics-aware triage](#). *IEEE Transactions on Medical Imaging*, 42(12):3579–3589.

Youngtaek Oh, Jae Won Cho, Dong-Jin Kim, In So Kweon, and Junmo Kim. 2024. [Preserving multi-modal capabilities of pre-trained VLMs for improving vision-linguistic compositionality](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 19060–19076. Association for Computational Linguistics.

Amit Peleg, Naman Deep Singh, and Matthias Hein. 2025. [Advancing compositional awareness in CLIP with efficient fine-tuning](#). In *Advances in Neural Information Processing Systems*, volume 38.

Hai X. Pham, David T. Hoffmann, Ricardo Guerrero, and Brais Martinez. 2026. [No hard negatives required: Concept centric learning leads to compositionality without degrading zero-shot capabilities of contrastive models](#). In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastri, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. [Learning transferable visual models from natural language supervision](#). In *Proceedings of the 38th International Conference on Machine Learning*, volume 139, pages 8748–8763. PMLR.

A Complete Policy Definitions

A batch has B unique image rows and T text columns, containing all selected source and hypothesis captions for its images. The relation matrix $R \in \{0, 1, 2, 3, 4\}^{B \times T}$ uses codes for unannotated, source, supported, contradicted, and neutral pairs, respectively. Relations apply only to their annotated image-text pair. An absent cross-image annotation is an eligible negative, not a neutral label.

Let J denote the candidate set for a row or column, $P \subseteq J$ its positive set, and P_{ij} the binary positive mask. Every contrastive policy includes source captions. Expanded policies additionally include supported hypotheses. Candidate weights start at one and are changed according to Table 4; all positive weights are then forced to one. The scalar loss averages eligible image rows and eligible text columns independently, with weight $1/2$ on each direction. The reverse direction transposes the same mask and weight matrices. Columns without a positive image are omitted from the reverse average. These rules preserve the meaning of rectangular batches with several captions per image.

Detached similarity weights. Let t_j be the unit frozen text feature for candidate j , and C_i the source-caption indices of image i . Define

$$a_{ij} = \frac{1 + |C_i|^{-1} \sum_{k \in C_i} t_k^\top t_j}{2}. \quad (3)$$

The average of source features is not renormalized. The negative weight is

$$w_{ij} = \text{sigmoid}\left(\frac{\tau - a_{ij}}{0.05}\right). \quad (4)$$

The fixed rule sets $\tau = 0.5$. The median rule uses the statistical median of a_{ij} over nonpositive cells of the current batch, averaging the two central values for an even count. Positive weights remain one. All features, thresholds, and weights in this calculation are detached from the training graph. The kernel never uses adapted features.

Shuffling and random exclusion. The shuffled rule permutes the complete batch-wide multiset of negative weights from the fixed-threshold rule, leaving positive weights intact. It preserves the weight distribution but need not preserve each row’s weighted mass. Random exclusion samples nonpositive cells without replacement within each image row, matching that image’s number of neutral annotations. Because the reverse direction uses

a transpose, its per-text exclusion counts need not match the neutral rule. Both randomizations use a seeded generator whose state advances through batches and epochs.

Smoothing and ranking. Smoothing uses source-positive targets and unit weights. Its row target is $0.9 \text{Unif}(P) + 0.1 \text{Unif}(J)$, where J contains every candidate. For ranking, let E_i and C'_i denote supported and contradicted hypotheses of image i . The eligible-image loss is

$$\ell_i^{\text{rank}} = \frac{\sum_{e \in E_i} \sum_{c \in C'_i} \text{softplus}(s_{ic} - s_{ie})}{|E_i| |C'_i|}. \quad (5)$$

We average over images with both sets nonempty. Ranking has no reverse direction and does not train on source, neutral, or unannotated pairs.

B Data, Provenance, and Reconstruction

The current execution was reconstructed from retained methods and proof text after the earlier workspace became unavailable. Historical aggregate tables are kept in a separate context directory and are not inputs to training, evaluation, or paper tables. The exact old similarity kernel and the base target set for smoothing were not recoverable. The explicit definitions in Appendix A were fixed before the new runs and are documented in the reconstruction record. The present study is a new execution under those recorded choices.

B.1 Relation annotations and source captions

We use the official e-ViL release of e-SNLI-VE at revision 94f17f5c1895. Its training, development, and test sources contain 401,717, 14,339, and 14,740 annotated hypotheses over 29,783, 1,000, and 1,000 images, respectively. The official pools have no shared image IDs or annotation-pair IDs. With Python’s `random.Random(42)`, we sample from sorted image IDs: 1,200 training images, 200 development images split into 100 calibration and 100 validation images, and 400 test images. All annotations of each selected image remain in source order. Table 5 gives the resulting counts.

The source captions come from the official Flickr30k Entities annotation archive at revision 68b3d6f12d1d. We use its five sentence-file lines per image in stored order. Entity markup is removed while retaining source words, case, and punctuation, with single spaces between stored tokens. This source covers all 31,783 images. An incomplete alternative caption mirror covers

Archive name	Positive set	Weight or target rule
clip	Source	All weights one
multipositive	Source + supported	All weights one
grounded	Source + supported	Contradictions: 2; neutrals: 0
grounded_no_hardening	Source + supported	Neutrals: 0
grounded_no_abstention	Source + supported	Contradictions: 2
random_exclusion	Source + supported	Count-matched random exclusions
smoothing	Source	$0.9 \text{ Unif}(P) + 0.1 \text{ Unif}(J)$
constant	Source	All negatives: 0.25
sanw_fixed	Source	Similarity kernel, threshold 0.5
sanw_median	Source	Similarity kernel, batch median threshold
shuffled	Source	Permuted fixed-threshold negative weights
pairwise_rank	Supported/contradicted	Image-balanced pairwise softplus

Table 4: All twelve trainable policies. A weight of zero exactly excludes a pair. The frozen reference has no trained parameters.

Split	Images	Support	Contradict	Neutral
Train	1,200	5,319	5,846	5,027
Calibration	100	522	554	355
Validation	100	502	534	339
Test	400	2,088	2,278	1,495

Table 5: Retained hypothesis annotations. Each image also has five source captions. All validation and test images, and 1,192 of 1,200 training images, have both supported and contradicted hypotheses.

only 31,014 and was not used. The image bytes come from the `nlphuji/flickr30k` mirror at revision `2b239befc81b`. Each selected image passes ZIP size/CRC checks, JPEG verification, and full decoding. No selected image is silently skipped.

An earlier deprecated relation release was rejected during an input-only audit because of development/test overlap. This decision preceded feature-based evaluation. The chosen official release, exact IDs, source URLs, full revisions, file hashes, and selection order are recorded in the per-dataset provenance files. The released labels include inherited text-inference judgments, so their names are operational annotation categories rather than a claim of new visual adjudication.

B.2 Transfer benchmarks

SugarCrepe uses all 7,511 pairs over 1,560 images from revision `0047054b2439`. Its category counts are 692 added attributes, 2,062 added objects, 788 replaced attributes, 1,652 replaced ob-

jects, 1,406 replaced relations, 666 swapped attributes, and 245 swapped objects. SugarCrepe++ uses all 4,757 triplets over 1,542 images from revision `c083bbf039a5`; it covers the five replacement/swap categories with the same category counts.

Each benchmark preserves its own released text strings. SugarCrepe++ changes 1,203 first-positive strings and 878 foil strings relative to the matched SugarCrepe records. Its first-positive accuracy therefore cannot substitute for SugarCrepe’s replacement/swap score. Two released SugarCrepe++ triplets have identical first-positive and foil strings, making a strict win impossible; these remain in the official denominator. One triplet repeats the same string in both positive positions. No item is edited or removed in response to model scores.

COCO uses all 5,000 Stanford Karpathy test images. The source contains 25,010 captions; taking the first five in stored order for every image yields 25,000 candidates and omits ten sixth captions. Caption strings are not normalized. The COCO JPEG cache contains 6,377 distinct benchmark images. It uses the official `val2014` source for Karpathy members and `val2017` for remaining benchmark images. All 183 shared SugarCrepe/Karpathy images were checked across these source versions and are byte-identical.

B.3 Overlap and duplicate audit

All 1,542 SugarCrepe++ images also occur in SugarCrepe. COCO retrieval shares 183 images with SugarCrepe and 182 with SugarCrepe++. These are complementary measurements on overlapping

pools. The selected Flickr30k images share neither IDs nor image-byte hashes with the COCO-derived benchmarks. The four e-SNLI-VE splits also have no shared image bytes.

The input audit finds 48 repeated exact image/text groups in e-SNLI-VE, each appearing twice. Of these, 46 are repeated hypothesis groups. Their relation labels do not conflict. All rows are retained, so repeated annotations retain their multiplicity in the loss and metrics. The two positive/foil equality cases account for the within-triplet relation conflicts in SugarCrepe++. The audit records source defects before scoring rather than changing benchmark contents.

B.4 Frozen feature extraction

The encoder is laion/CLIP-ViT-B-32-laion2B-s34B-b79K, revision 1a25a446712b, with weight SHA256 beginning ac4f8c4b88af. We use OpenCLIP 2.32.0 and PyTorch 2.7.1 on CPU. Images are resized with bicubic interpolation, center-cropped to 224×224 , converted to RGB, and normalized with the checkpoint’s channel means and standard deviations. The 512-dimensional image and text outputs are stored in float32 after unit L2 normalization. The fixed scale is 100.00000762939453.

Text encoding trims only causal padding positions after the last end-of-text token in a batch. A parity check compares normalized outputs against the stock encoder with tolerance $2 \cdot 10^{-6}$. Recorded per-dataset parity checks, preprocessing parameters, source hashes, input-manifest hashes, feature-cache hashes, and full model pins are stored in each feature metadata file. Reused features require exact image-byte or text-string identity under the same encoder and preprocessing.

C Training and Checkpoint Selection

Let $x, z \in \mathbb{R}^{512}$ be unit frozen image and text features. The trainable maps are

$$\hat{x} = \frac{x + A_I x}{\|x + A_I x\|_2}, \quad \hat{z} = \frac{z + A_T z}{\|z + A_T z\|_2}. \quad (6)$$

Both 512×512 matrices start at zero and have no bias. Their total parameter count is $2 \cdot 512^2 = 524,288$. The score is $s_{ij} = \alpha \hat{x}_i^\top \hat{z}_j$, where $\alpha > 0$ is the pretrained checkpoint’s fixed scale. Feature extraction and source hashes identify the exact encoder and input order.

The learning-rate grid is $\{10^{-4}, 3 \cdot 10^{-4}, 10^{-3}\}$ and training seeds are 17, 29, 43. Each candidate

trains for ten epochs with image batches of 32. All source captions and hypotheses assigned to those images enter the text candidate pool. AdamW uses weight decay 0.01, betas (0.9, 0.999), and epsilon 10^{-8} . The gradient norm is clipped at one. There is no warmup or learning-rate schedule. Separate random generators control image order and stochastic policy operations, so weight shuffling does not alter batch order.

Validation retrieval uses every validation caption and hypothesis as a candidate. Both source captions and supported hypotheses are relevant matches. Its R@1 and the image-balanced supported-versus-contradicted accuracy receive equal weight in the selection score. The epoch with the highest score is retained for each seed and learning rate, including epoch zero. Exact ties retain the earliest epoch. A policy selects one learning rate by averaging these best scores over all three seeds; exact ties select the smaller rate. Held-out scoring starts after all 108 candidates and 36 selections are frozen.

Ranking evaluation accumulates dot products in float64 over adapted float32 normalized features. Exact retrieval-score ties follow candidate manifest order; relation-ranking ties receive half credit. The fixed positive score scale is omitted from ranking metrics because it cannot change the order. Images without both supported and contradicted hypotheses are omitted only from the relation average, with their count recorded.

An immutable execution ledger binds the protocol, code, manifest, feature metadata, cache, and split IDs before fitting. Every candidate saves its best state, all eleven validation evaluations, and a completion record with artifact hashes. The root epoch-zero state is retained separately. Resume accepts a completed candidate only when its identities and hashes match. Interrupted candidates restart deterministically from epoch zero. Selected checkpoints are copied into a separate tree while all candidate states remain available.

D Metrics and Statistical Comparisons

For e-SNLI-VE, compare every supported hypothesis against every contradicted hypothesis within an image. A larger supported score receives one point and an exact tie receives half a point. Average pairs within each eligible image, then average images. This gives each image equal weight despite different numbers of annotated statements.

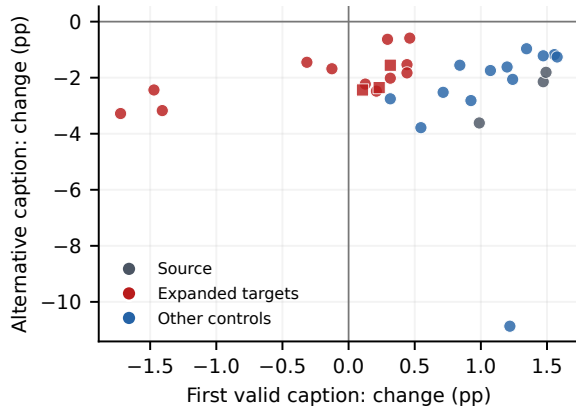


Figure 2: Changes on the same SugarCrepe++ triplets relative to frozen. Each point is one selected nonzero-epoch adapter. All 31 lose accuracy on the alternative valid caption; 26 improve on the first. The 5 epoch-zero selections are omitted.

SugarCrepe uses a strict positive-score comparison against its foil. SugarCrepe++ reports each positive separately and their conjunction against the same foil. The conjunction is one only if both scores are strictly greater. Per-category results retain official labels. The addition group pools added objects and attributes; the combined replacement/swap group pools the remaining categories. Grouped scores use item counts as weights, not an unweighted mean of category percentages.

COCO image-to-text R@1 counts an image query as correct when its highest-scoring caption belongs to the image’s annotated caption set. Text-to-image R@1 requires the source image at rank one. All candidates remain in the full pool during scoring. Source-caption ownership is the operational relevance label.

The six primary contrasts compare the combined relation policy against positive expansion, neutral exclusion alone, and contradiction weighting alone, on e-SNLI-VE and SugarCrepe. For each contrast, average the paired outcomes across the three selected seeds before sampling images. Each bootstrap draw samples images with replacement and retains all associated items. With 10,000 replicates, seed 20261003, and family size six, two-sided percentile bounds use tail probability $0.05/(2 \cdot 6)$, yielding 99.1667% intervals. They condition on the selected runs and data images. Secondary metrics and category groupings are descriptive.

E Retention Across Selected Adapters

The first-positive and alternative-positive scores use the same triplets and foil. The decline in alternative-caption accuracy is relative to frozen and is not specific to positive expansion. Positive expansion has slightly higher alternative-caption accuracy than source-positive adaptation. Full per-policy and per-seed values follow below.

F Full Fresh Results and Execution Checks

Tables 6–13 give all selected-policy means, seed outcomes, primary intervals, category scores, and retrieval transitions. Five selected states retain epoch zero: all three ranking runs and fixed-similarity seeds 29 and 43. Those states have predictions identical to the frozen reference.

The illustrative query in Figure 1 is coco 60623. It is the first image in COCO manifest order that source-positive adaptation retrieves correctly at rank one in all three seeds, while positive expansion fails at rank one in every seed and falls outside the top five in seed 17. The displayed captions are the two seed-17 top results. This deterministic post hoc choice illustrates a recorded failure and is not a representative sample or an extra significance test.

G Separate Relation-Estimator Diagnostic

This diagnostic predicts released supported, contradicted, and neutral labels from frozen features. It does not provide labels or weights to the adapter study. Source captions and unannotated cross-image pairs are excluded. Each annotated image/hypothesis pair is one example, and each image remains entirely within its training, validation, calibration, or test split.

Features and fitting. The image-only and text-only models use the corresponding 512-dimensional features. The multimodal model concatenates image, text, their elementwise product, and their absolute difference. A standard scaler is fitted only on training rows. Multinomial logistic regression uses an intercept, L2 regularization, `lbfgs`, tolerance 10^{-6} , no class weights, and seed 17. The grid $C \in \{0.01, 0.1, 1, 10\}$ is selected by validation macro-F1, with smaller C breaking ties. There is no refit on validation data.

Policy	e-SNLI-VE	SugarCrepe	SC++ both
Frozen	79.47	81.21	68.95
Source	81.18 $\pm$ 0.14	83.26 $\pm$ 0.30	67.40 $\pm$ 0.84
Expanded	87.00 $\pm$ 0.40	83.39 $\pm$ 0.15	67.23 $\pm$ 0.34
Contradiction	87.28 $\pm$ 0.33	83.33 $\pm$ 0.16	67.30 $\pm$ 0.28
Neutral	86.66 $\pm$ 0.44	83.39 $\pm$ 0.27	67.66 $\pm$ 0.52
Combined	86.74 $\pm$ 0.17	83.24 $\pm$ 0.31	67.61 $\pm$ 0.45
Random	86.26 $\pm$ 0.28	81.62 $\pm$ 0.70	65.93 $\pm$ 0.57
Smoothing	79.23 $\pm$ 4.85	82.28 $\pm$ 2.18	65.97 $\pm$ 4.59
Ranking	79.47 $\pm$ 0.00	81.21 $\pm$ 0.00	68.95 $\pm$ 0.00
Fixed	79.56 $\pm$ 0.15	81.80 $\pm$ 1.01	68.00 $\pm$ 1.65
Shuffled	83.34 $\pm$ 0.57	83.49 $\pm$ 0.36	66.90 $\pm$ 0.26
Median	82.66 $\pm$ 0.19	83.73 $\pm$ 0.17	68.01 $\pm$ 0.07
Constant	82.15 $\pm$ 0.24	83.67 $\pm$ 0.20	68.37 $\pm$ 0.48

Table 6: All selected policies on the three discrimination endpoints. Entries are percentages with sample seed standard deviations. Source, Expanded, Contradiction, Neutral, and Combined correspond to the first five trainable policies in Table 1. Other names follow Table 4.

The original reconstruction set a 2,000-iteration cap. Text fitting reached that cap, so a separately recorded amendment raised the cap to 10,000 without changing the model, grid, splits, or tolerance. The amended run has its own ledger and output directory; the original diagnostic remains archived. This numerical-completion amendment does not alter the frozen primary-adaptor protocol.

Calibration and selection gates. Sort calibration image IDs and permute them with NumPy seed zero. The first 50 images fit a positive scalar temperature by minimizing mean three-class negative log likelihood; the other 50 select gates. Temperature lies in $[0.05, 20]$, optimized in log space, with one and both endpoints also considered explicitly. It preserves the classifier’s argmax predictions.

For supported and contradicted predictions separately, a gate requires that class to be the argmax and its calibrated probability to meet or exceed a threshold. Search complete confidence-tie groups and choose maximum coverage among thresholds accepting at least 20 calibration pairs with a two-sided 95% Wilson lower endpoint of at least 0.90. If no threshold qualifies, the gate accepts nothing. This calibration criterion is not a simultaneous guarantee after threshold search. Final precision and coverage are measured on held-out test images. Coverage divides accepted count by all test hypotheses; precision divides correct accepted predictions by accepted count. An empty gate has undefined precision.

Metrics and out-of-fold predictions. The diagnostic reports accuracy, three-class macro-F1, 10-bin expected calibration error, Brier score, and negative log likelihood. Brier score sums squared error over the three classes before averaging examples. Calibration bins have equal width and include probability one in the last bin. Class ties follow the fixed order supported, contradicted, neutral. Nonempty gate precision receives a descriptive 95% image-bootstrap interval from 10,000 replicates with seed 20261003. Empty-acceptance replicates are counted and omitted from the percentile calculation.

For reusable training predictions, sort training image IDs, permute with seed 17, and divide them into three image-disjoint folds. Each fold trains on the other two folds, selects C' on the external validation split, and calibrates on the external calibration halves. Thus held-out training labels do not enter the model or selection decisions producing their predictions. These predictions are archived for future work and do not enter adaptor fitting.

H Reproducibility and Extensions

The archive separates source acquisition, normalized feature extraction, fit/selection, held-out prediction, statistical analysis, and manuscript generation. Manifests give stable IDs and input hashes. Candidate checkpoints and validation histories establish which states were eligible for selection. Selected predictions support metric reconstruction without rerunning training. Result tables are gen-

Policy	SC++ first	SC++ alternative	COCO I→T	COCO T→I
Frozen	79.46	75.01	56.32	39.35
Source	80.78 $\pm$ 0.29	72.48 $\pm$ 0.96	55.85 $\pm$ 0.53	39.38 $\pm$ 0.32
Expanded	79.68 $\pm$ 0.11	72.89 $\pm$ 0.49	47.29 $\pm$ 1.78	34.84 $\pm$ 0.89
Contradiction	79.72 $\pm$ 0.16	72.92 $\pm$ 0.49	47.25 $\pm$ 1.94	34.81 $\pm$ 0.92
Neutral	79.72 $\pm$ 0.33	73.64 $\pm$ 0.68	48.01 $\pm$ 0.15	35.39 $\pm$ 0.23
Combined	79.56 $\pm$ 0.36	73.64 $\pm$ 0.70	47.26 $\pm$ 1.17	35.27 $\pm$ 0.44
Random	77.93 $\pm$ 0.17	72.04 $\pm$ 0.46	43.43 $\pm$ 1.23	29.95 $\pm$ 0.65
Smoothing	80.88 $\pm$ 0.18	70.56 $\pm$ 5.56	53.85 $\pm$ 2.54	37.08 $\pm$ 3.15
Ranking	79.46 $\pm$ 0.00	75.01 $\pm$ 0.00	56.32 $\pm$ 0.00	39.35 $\pm$ 0.00
Fixed	79.64 $\pm$ 0.32	73.74 $\pm$ 2.18	54.33 $\pm$ 3.45	36.86 $\pm$ 4.31
Shuffled	80.11 $\pm$ 0.31	72.31 $\pm$ 0.15	52.27 $\pm$ 0.29	38.19 $\pm$ 0.12
Median	80.50 $\pm$ 0.18	73.37 $\pm$ 0.10	53.97 $\pm$ 0.40	39.21 $\pm$ 0.27
Constant	80.84 $\pm$ 0.16	73.60 $\pm$ 0.58	55.21 $\pm$ 0.14	39.39 $\pm$ 0.09

Table 7: Valid-caption and retrieval retention. Entries are percentages with sample seed standard deviations.

erated from the fresh analysis artifact rather than entered from the historical manuscript.

Extensions retained for future work include a second backbone, full-encoder training, more training data and seeds, independently image-adjudicated relations, shared-final-epoch contrasts, denominator-matched and exact gradient-mass-matched training controls, hardness-matched exclusion, the summed-positive objective, predicted relations, inverse-multiplicity weighting, retention-constrained selection, feature distillation, many-to-many retrieval relevance, natural deployment queries, and frozen retrieval followed by relation-adaptor reranking. A reranker can change the top result even when its candidate generator is fixed. Matching controls must specify the image and text directions separately. These are distinct from the completed comparisons reported in the fresh results.

I Analytical Identities and Their Scope

This appendix collects the positive-target identity in the main text and the complete negative-weight analysis. Unless an alternative is specified, the statements concern the unsmoothed uniform-positive objective at specified scores. Their proofs do not depend on experimental outcomes.

I.1 Notation and the common gradient

Let J be a finite candidate set, $P \subseteq J$ a nonempty positive set, and $N = J \setminus P$. Each score $s_j \in \mathbb{R}$ is finite. Weights $w_j \geq 0$ are fixed for differentiation

and satisfy $w_p = 1$ for $p \in P$. Define

$$D = \sum_{j \in J} w_j e^{s_j}, \quad q_j = \frac{w_j e^{s_j}}{D}, \quad (7)$$

$$y_j = \frac{\mathbf{1}\{j \in P\}}{|P|}, \quad \ell = \log D - \sum_j y_j s_j. \quad (8)$$

Since P is nonempty and scores are finite, $D > 0$. A zero weight removes a candidate from the denominator, while its target is also zero.

Lemma 1 (Fixed-weight row gradient). *For each $j \in J$, $\partial \ell / \partial s_j = q_j - y_j$.*

Proof. Differentiating the finite sum gives $\partial D / \partial s_j = w_j e^{s_j}$. The logarithm contributes $w_j e^{s_j} / D = q_j$, and the linear target term contributes $-y_j$. $\square$

An individual positive derivative need not be negative. If $P = \{1, 2\}$ and $q = (0.8, 0.1, 0.1)$, the gradient is $(0.3, -0.4, 0.1)$. However,

$$\sum_{p \in P} \frac{\partial \ell}{\partial s_p} = - \sum_{n \in N} q_n \leq 0. \quad (9)$$

Thus claims about total positive attraction do not imply a sign for every positive score.

I.2 Expansion of the positive set

Proposition 1 (Target redistribution). *Assume unit weights on every candidate and fixed scores. Let P contain $K \geq 1$ positives. Relabel a disjoint set $A \subseteq J \setminus P$ of $M \geq 0$ already-present candidates as positive, yielding $P' = P \cup A$. For $p \in P$,*

$$\frac{\partial \ell'}{\partial s_p} - \frac{\partial \ell}{\partial s_p} = \frac{M}{K(K+M)}. \quad (10)$$

Policy	Seed	Rate	Epoch	e-SNLI	Sugar	SC++	I→T	T→I
Source	17	0.0001	1	81.35	83.44	67.65	56.28	39.65
Source	29	0.0001	1	81.09	83.42	68.09	56.02	39.46
Source	43	0.0001	3	81.12	82.91	66.47	55.26	39.02
Expanded	17	0.0003	5	86.75	83.38	67.58	48.36	35.40
Expanded	29	0.0003	10	86.79	83.54	66.91	45.24	33.82
Expanded	43	0.0003	5	87.46	83.25	67.19	48.28	35.30
Contradiction	17	0.0003	5	87.04	83.33	67.58	48.46	35.48
Contradiction	29	0.0003	10	87.13	83.49	67.02	45.02	33.76
Contradiction	43	0.0003	5	87.66	83.17	67.31	48.28	35.18
Neutral	17	0.0003	5	86.42	83.60	68.26	48.10	35.54
Neutral	29	0.0003	5	86.40	83.49	67.35	47.84	35.50
Neutral	43	0.0003	5	87.17	83.09	67.37	48.10	35.12
Combined	17	0.0003	5	86.64	83.46	68.09	48.04	35.53
Combined	29	0.0003	5	86.65	83.37	67.19	47.82	35.51
Combined	43	0.0003	7	86.94	82.88	67.56	45.92	34.77
Random	17	0.001	3	86.04	80.96	65.48	44.84	29.33
Random	29	0.001	5	86.17	81.55	66.58	42.80	30.62
Random	43	0.001	3	86.58	82.36	65.73	42.64	29.90

Table 8: Individual selected checkpoints. SC++ requires both positives to beat the foil; retrieval columns are full-pool R@1. All metric entries are percentages.

For $a \in A$, the difference is $-1/(K + M)$. For the remaining candidates, it is zero.

Proof. The scores, candidates, and weights are unchanged, so D and all q_j remain fixed. For an original positive, subtracting the two derivatives gives

$$(q_p - 1/(K + M)) - (q_p - 1/K) = \frac{M}{K(K + M)}. \quad (11)$$

A newly positive candidate changes target probability from zero to $1/(K + M)$. The remaining targets stay zero. The original positives jointly lose $M/(K + M)$ target probability, precisely the amount assigned to the new positives. $\square$

The premise that candidates are already present matters: adding new candidates changes the denominator as well. Likewise, changing weights changes q . The following identity separately accounts for a change in the eligible text anchors of symmetric training.

Reverse-anchor averaging. Hold the candidate images, caption columns, scores, and unit weights fixed. Let C_0 be the text columns with a positive image before expansion and H the additional eligible columns afterward. Assume that the new eligible set is the disjoint union $C_0 \cup H$ and that

positive sets for columns in C_0 are unchanged. This holds for source-caption columns when supported hypothesis columns acquire their first positive image. Define $T_0 = |C_0| > 0$, and let L_{rev} be the original mean text-to-image loss. Then

$$L'_{\text{rev}} = \frac{T_0 L_{\text{rev}} + \sum_{j \in H} \ell'_j}{T_0 + |H|}. \quad (12)$$

Every old column retains its loss. Splitting the new average into old and new columns proves the identity. It also holds for empty H . Original text anchors retain fraction $T_0/(T_0 + |H|)$ of the reverse average. This change in anchor averaging is separate from target redistribution within an image row. The complete symmetric loss gives each direction weight $1/2$.

I.3 A constant matching the normalizer

Proposition 2 (Matched negative mass). *Let N be nonempty under the common assumptions. Define*

$$c^* = \frac{\sum_{n \in N} w_n e^{s_n}}{\sum_{n \in N} e^{s_n}}. \quad (13)$$

Compute c^ at the specified scores and hold it fixed for differentiation. Replace w_n by this constant for each $n \in N$, keeping positive weights one. At*

Policy	Seed	Rate	Epoch	e-SNLI	Sugar	SC++	I→T	T→I
Smoothing	17	0.0001	1	82.04	83.61	68.59	55.26	38.99
Smoothing	29	0.0001	10	73.63	79.76	60.67	50.92	33.45
Smoothing	43	0.0001	1	82.03	83.48	68.64	55.36	38.81
Ranking	17	0.0001	0	79.47	81.21	68.95	56.32	39.35
Ranking	29	0.0001	0	79.47	81.21	68.95	56.32	39.35
Ranking	43	0.0001	0	79.47	81.21	68.95	56.32	39.35
Fixed	17	0.0003	6	79.74	82.97	66.09	50.34	31.88
Fixed	29	0.0003	0	79.47	81.21	68.95	56.32	39.35
Fixed	43	0.0003	0	79.47	81.21	68.95	56.32	39.35
Shuffled	17	0.0003	1	82.85	83.10	66.64	52.60	38.10
Shuffled	29	0.0003	1	83.97	83.81	66.89	52.14	38.14
Shuffled	43	0.0003	1	83.20	83.54	67.16	52.06	38.33
Median	17	0.0001	3	82.76	83.54	67.96	53.64	38.90
Median	29	0.0001	2	82.44	83.74	67.96	54.42	39.42
Median	43	0.0001	2	82.78	83.89	68.09	53.86	39.30
Constant	17	0.0001	1	82.00	83.88	68.59	55.36	39.48
Constant	29	0.0001	1	82.03	83.65	68.70	55.10	39.39
Constant	43	0.0001	2	82.43	83.48	67.82	55.16	39.31

Table 9: Individual selected checkpoints. SC++ requires both positives to beat the foil; retrieval columns are full-pool R@1. All metric entries are percentages.

Comparator	Endpoint	Δ	99.17% interval	Seed 17	Seed 29	Seed 43
Expanded	e-SNLI-VE	-0.255	[-0.605, +0.065]	-0.109	-0.137	-0.520
Expanded	SugarCrepe	-0.155	[-0.509, +0.209]	+0.080	-0.173	-0.373
Neutral	e-SNLI-VE	+0.081	[-0.100, +0.273]	+0.221	+0.253	-0.230
Neutral	SugarCrepe	-0.155	[-0.371, +0.056]	-0.133	-0.120	-0.213
Contradiction	e-SNLI-VE	-0.535	[-0.890, -0.193]	-0.398	-0.484	-0.723
Contradiction	SugarCrepe	-0.093	[-0.442, +0.261]	+0.133	-0.120	-0.293

Table 10: Six primary contrasts: Combined minus the comparator, in percentage points. Intervals use paired image-cluster resampling and the fixed six-comparison Bonferroni family.

these scores, the new normalizer, loss, each positive probability and derivative, and total negative derivative equal their original values.

Proof. The denominator defining c^* is positive because N is nonempty and every exponential is positive. Therefore c^* exists, is nonnegative, and satisfies

$$\sum_{n \in N} c^* e^{s_n} = \sum_{n \in N} w_n e^{s_n}. \quad (14)$$

The positive exponential mass is unchanged. Adding it to both sides proves equality of the normalizers. The target term also stays fixed, giving equal losses. Each positive probability is e^{s_p}/D , so its probability and derivative are equal. Finally, the total negative derivative is the negative exponential mass divided by D in each case. $\square$

Individual negative derivatives are generally unequal: they are $w_n e^{s_n}/D$ and $c^* e^{s_n}/D$. Matching preserves the total amount of competition, while changing its allocation. This is a row-specific analytical construction; a single global constant need not match every row. If instead $c^*(s)$ is recomputed differentiably, the new normalizer equals the original normalizer as a function of scores. Differentiating that equality restores the original individual negative derivatives. The allocation comparison therefore requires detachment.

The transpose need not preserve the match. Row-specific matching does not generally preserve the normalizers of the transposed text-to-image objective. For example, take two images, three captions, all scores zero, positive pairs (1, 1) and

Policy	Add A	Add O	Repl. A	Repl. O	Repl. R	Swap A	Swap O
Frozen	77.89	87.15	82.49	93.77	68.92	67.42	60.00
Source	83.24	89.72	83.76	94.09	72.00	67.47	61.77
Expanded	87.86	90.74	82.91	93.02	72.00	65.12	60.68
Contradiction	88.05	90.59	83.04	92.88	72.02	64.96	60.27
Neutral	87.91	90.98	82.91	93.18	71.46	65.17	60.41
Combined	87.91	90.74	83.04	93.20	71.08	65.27	59.05
Random	86.85	89.19	81.09	92.39	68.66	63.06	57.14
Smoothing	80.54	87.20	83.46	93.48	72.43	67.62	62.99
Ranking	77.89	87.15	82.49	93.77	68.92	67.42	60.00
Fixed	80.11	87.86	82.66	93.79	69.77	66.97	61.36
Shuffled	86.37	90.88	83.08	93.68	70.63	68.07	61.36
Median	85.12	91.21	83.71	94.05	71.03	68.22	62.31
Constant	84.20	90.80	83.46	94.23	72.00	67.57	62.31

Table 11: SugarCrep category means (percent). A, O, and R denote attribute, object, and relation. Category denominators are listed in Appendix B.

Seed	Queries	Pool	Lost	New	Net loss
17	I→T	5,000	649	253	396
17	T→I	25,000	2,195	1,132	1,063
29	I→T	5,000	832	293	539
29	T→I	25,000	2,651	1,241	1,410
43	I→T	5,000	597	248	349
43	T→I	25,000	1,976	1,045	931

Table 12: Retrieval success transitions from source positives to positive expansion. Lost and new count changed rank-one successes on the identical query pool.

(2, 2), and weights

$$W = \begin{pmatrix} 1 & 0.2 & 0.8 \\ 0.6 & 1 & 0.4 \end{pmatrix}. \quad (15)$$

Both row constants are 0.5 and both image-row losses remain $\log 2$. The two eligible text-column normalizers change from 1.6, 1.2 to 1.5, 1.5. Consequently the full symmetric loss increases by

$$\frac{1}{4} \log \frac{1.5^2}{1.6 \cdot 1.2} \approx 0.0396513. \quad (16)$$

A symmetric matched control therefore requires additional constraints. The executed fixed constant 0.25 is not a c^* -matched experiment.

I.4 Suppression does not control every negative group

For one positive, let all negative weights lie in $[0, 1]$. Write

$$S = \sum_j e^{s_j}, \quad D = e^{s_+} + \sum_{n \in N} w_n e^{s_n}. \quad (17)$$

The inequality $D \leq S$ implies $q_+ = e^{s_+}/D \geq p_+ = e^{s_+}/S$. Hence total negative derivative satisfies

$$\sum_{n \in N} q_n = 1 - q_+ \leq 1 - p_+. \quad (18)$$

This bound does not hold for every subset of negatives. Consider exponential masses 1 for the positive, 1 for a chosen negative group, and 100 for other negatives. Weight the chosen group by 0.5 and the others by 0.001. The chosen group’s probability changes from

$$\frac{1}{102} \approx 0.009804 \quad \text{to} \quad \frac{0.5}{1 + 0.5 + 0.1} = 0.3125. \quad (19)$$

The total negative mass decreases, but that group’s mass increases. Contradiction weighting by two also lies outside the premise $w_n \leq 1$.

I.5 Summed-positive probability

Let $A = \sum_{p \in P} e^{s_p}$. A different objective is

$$\ell_{\text{set}} = \log D - \log A. \quad (20)$$

Policy	I→T R1	R5	R10	T→I R1	R5	R10
Frozen	56.32	79.82	87.12	39.35	65.40	75.62
Source	55.85	79.59	86.93	39.38	65.67	75.91
Expanded	47.29	71.56	80.89	34.84	60.77	71.50
Contradiction	47.25	71.61	80.87	34.81	60.78	71.46
Neutral	48.01	72.58	81.79	35.39	61.56	72.19
Combined	47.26	71.93	81.33	35.27	61.36	72.05
Random	43.43	67.84	77.81	29.95	55.28	66.61
Smoothing	53.85	77.94	85.71	37.08	63.47	73.72
Ranking	56.32	79.82	87.12	39.35	65.40	75.62
Fixed	54.33	77.83	85.75	36.86	62.66	73.19
Shuffled	52.27	76.63	84.83	38.19	64.41	74.58
Median	53.97	77.93	85.83	39.21	65.42	75.66
Constant	55.21	78.85	86.55	39.39	65.62	75.85

Table 13: Full-pool COCO recall at ranks one, five, and ten, averaged across selected seeds (percent).

Input	Probabilities	Acc.	Macro-F1	ECE	Brier	NLL
Image	uncalibrated	35.64	34.06	4.73	0.674	1.111
Image	calibrated	35.64	34.06	0.05	0.666	1.098
Text	uncalibrated	61.76	61.38	5.78	0.509	0.879
Text	calibrated	61.76	61.38	0.68	0.503	0.858
Multimodal	uncalibrated	65.69	64.96	12.53	0.487	0.887
Multimodal	calibrated	65.69	64.96	1.38	0.457	0.783

Table 14: Held-out relation classification on the newly sampled test set. Accuracy, macro-F1, and ECE are percentages; Brier and NLL retain their native scale.

For $p \in P$, direct differentiation gives

$$\frac{\partial \ell_{\text{set}}}{\partial s_p} = q_p - \frac{e^{s_p}}{A} \leq 0, \quad (21)$$

because $D \geq A$. The attraction is allocated according to the positive scores, rather than equally across positives. This permits the highest-scoring positives to dominate the target distribution. The identity describes a distinct objective, not an equivalent implementation of Equation 1.

I.6 Parameter-dependent weights and score updates

Let J_+ be a fixed nonzero-support set containing P . For $j \in J_+$, assume that $w_j(\theta) > 0$ is locally differentiable, with $w_p(\theta) \equiv 1$ on positives. All other weights are fixed at zero. With fixed targets and differentiable parameter-dependent scores,

$$\begin{aligned} \nabla_{\theta} \ell &= \sum_{j \in J_+} (q_j - y_j) \nabla_{\theta} s_j \\ &+ \sum_{j \in J_+} q_j \nabla_{\theta} \log w_j. \end{aligned} \quad (22)$$

The second term vanishes for detached weights. No logarithmic derivative is taken at a zero weight; the identity does not describe a change of support through zero. Even with detached weights, a parameter update combines the score Jacobians across every candidate and batch. A sign or magnitude bound for an individual score derivative is not a corresponding bound on the parameter-gradient norm or on the score after optimization.

I.7 Other fixed-score identities

The following facts collect the single-positive special cases behind similarity suppression. Let $0 \leq w_n \leq 1$ for negatives, $w_+ = 1$, and

$$R = \sum_{n \in N} (1 - w_n) e^{s_n}, \quad D = S - R. \quad (23)$$

Since $0 \leq R < S$,

$$\log q_+ - \log p_+ = -\log(1 - R/S) \geq R/S \geq 0. \quad (24)$$

The logarithm inequality follows by integrating $1/(1-x) \geq 1$ over $[0, R/S]$. It compares probabil-

Input	Gate	Accept	Correct	Coverage	Precision	95% interval
Image	supported	0	0	0.00	n/a	n/a
Image	contradicted	0	0	0.00	n/a	n/a
Text	supported	0	0	0.00	n/a	n/a
Text	contradicted	0	0	0.00	n/a	n/a
Multimodal	supported	0	0	0.00	n/a	n/a
Multimodal	contradicted	683	647	11.65	94.73	[92.72, 96.59]

Table 15: Selective test predictions. Precision is undefined when no pair is accepted. Intervals resample whole test images and are descriptive.

Input	C	Temp.	Max. iter.	OOF F1
Image	0.1	3.085	1182	33.17
Text	1	1.312	2318	63.46
Multimodal	0.01	1.904	2505	67.22

Table 16: Classifier selection and completion. Maximum iteration count is over the full-training grid; OOF is image-disjoint out-of-fold macro-F1 (percent).

ities at fixed scores and does not change their raw ranking. For a group of m negatives at common score a and common weight w , its gradient mass is

$$G = \frac{mwe^a}{D}. \quad (25)$$

The denominator also depends on the group. Let A be the unweighted exponential mass outside that group, so $S = A + me^a$. Nonnegative suppression elsewhere gives

$$R \geq m(1-w)e^a, \quad \frac{R}{S} \geq \frac{m(1-w)e^a}{A + me^a}. \quad (26)$$

This rational lower bound tends to $1-w$ as m grows with A, a, w fixed. The actual log-probability gain need not saturate; with $w = 0$ and no other suppressed candidates, it can grow without bound as m increases. At equal scores, a smaller weight gives smaller normalized probability within the same denominator. This says nothing about whether the weighting identifies a correct semantic relation.

As every weight tends to one, the weighted loss converges to the corresponding unweighted uniform-positive loss by continuity. A common negative weight below one does not yield the ordinary loss while the positive weight remains one. Finally, weighted normalization is ordinary softmax normalization on shifted scores $s_j + \log w_j$, with $\log 0 = -\infty$. It is not label smoothing, which changes the targets.

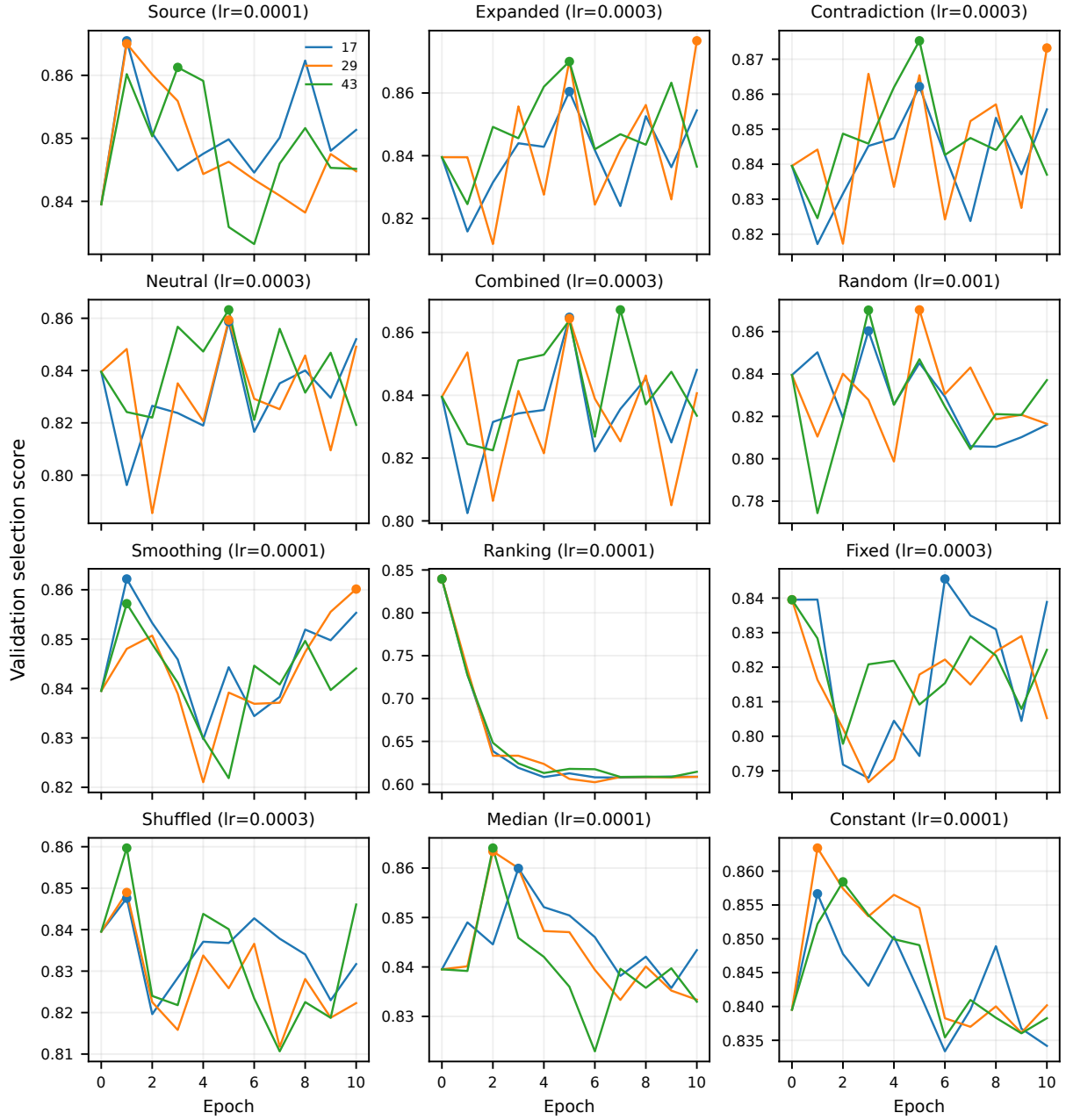


Figure 3: Validation trajectories at the selected learning rate of each policy. Lines are seeds; dots mark retained epochs. Every candidate evaluated epoch zero through ten. These are selection data, not held-out outcomes.